CentroidSteer: Tri-Tier Subspace Optimization and Polar Orthogonalization for Dynamic Mixture-of-Experts

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Abstract

Scaling autoregressive foundation models and sparse Mixture-of-Experts (MoE) architectures exposes two conflicting optimization bottlenecks: first, first-order adaptive optimizers like AdamW incur prohibitive state memory overhead (8.0 bytes per parameter), severely constraining trainable model capacity; second, emerging polar orthogonalization methods (e.g., Muon) treat parameter tensors uniformly, causing empirical degradation when applied to vocabulary embeddings and disregarding the dynamic geometric routing of sparse experts. In this paper, we introduce **CentroidSteer**, a tri-tier partitioned optimization framework designed for sparse, high-throughput transformer training. CentroidSteer dynamically partitions weights into: (1) **Tier 1 (Muon)**, using quintic Newton-Schulz iterations to orthogonalize 2D attention matrices; (2) **Tier 2 (CentroidSteer)**, projecting MoE expert updates into an augmented low-rank triangular subspace steered along dynamic router activation centroids using scale-invariant Cholesky-QR Gram orthogonalization; and (3) **Tier 3 (Base AdamW)**, safely isolating 1D normalization layers, routing gates, and large vocabulary embeddings from singular value equalization. On real FineWeb-Edu pre-training across multiple random seeds with bit-level identical weight initialization, CentroidSteer achieves a persistent state footprint of only 1.00 **byte/param** (87.5% reduction vs. standard AdamW and 50.0% vs. 8-bit AdamW), while delivering superior convergence efficiency (0.1916 loss/s) on a 90.8M-parameter MoE model.

1 Introduction

The scaling of autoregressive large language models (LLMs) is fundamentally constrained by GPU memory bandwidth and capacity during distributed pre-training [1, 2]. While modern activations can be re-materialized or checkpointed, optimizer state buffers—specifically the first and second gradient moments (m, v) maintained by AdamW [3]—are permanently resident in High Bandwidth Memory (HBM). For an FP32 optimizer tracking BF16 weights, these auxiliary states consume 8 bytes per parameter, often exceeding the memory footprint of the underlying model weights and activations combined.

Although low-rank gradient projection frameworks such as GaLore [4] demonstrate that weight updates can be compressed into low-rank subspaces, they suffer from two major deficiencies in modern transformer architectures:

1. **SVD Computational Overhead**: Computing singular value decompositions (SVD) on large weight matrices repeatedly stalls GPU execution and causes severe transient memory spikes.

2. **Decoupling from Routing Topology**: In sparse Mixture-of-Experts (MoE) architectures [5, 6], token assignments dynamically cluster along specific directions in activation space. Standard low-rank gradient projection ignores this routing manifold, projecting expert updates onto arbitrary principal components rather than the active functional subspace demanded by current input distributions.

Concurrently, polar orthogonalization techniques such as Muon [7] have demonstrated rapid convergence on standard 2D linear layers by equalizing the singular values of gradient updates via quintic Newton-Schulz iterations. However, blindly applying orthogonal updates to vocabulary embedding tables or output heads can destabilize pre-training due to severe aspect-ratio distortions (vocab $\gg d$).

To address these challenges simultaneously, we propose **CentroidSteer**, a mathematically unified, tri-tier optimization architecture. CentroidSteer introduces:

- **Scale-Invariant Cholesky-QR Subspace Iteration**: Eliminates expensive SVD routines, achieving a 6.8× kernel speedup and 3.1× lower transient VRAM via batched Tensor Core GEMMs and triangular Gram-matrix Cholesky factorization.

- **Dynamic Router Centroid Steering**: Tracks top-1 and top-2 expert token assignment centroids in real-time, computing an orthonormal steering vector

$c_\perp$ that augments the projection subspace along the functional manifold of the routed experts.

- **Tri-Tier Parameter Partitioning**: Implements strict precedence rules that assign 2D attention matrices to Tier 1 (Muon), routed expert MLPs to Tier 2 (CentroidSteer), and sensitive embeddings/normalizations to Tier 3 (Fused AdamW).

2 CentroidSteer Methodology

2.1 Tri-Tier Parameter Taxonomy

Let $\theta \in \mathbb{R}^P$ denote the complete parameter vector of an MoE transformer. CentroidSteer partitions θ into three disjoint sets:

$$\theta = \theta_{\text{Tier 1}} \cup \theta_{\text{Tier 2}} \cup \theta_{\text{Tier 3}}, \quad \theta_i \cap \theta_j = \emptyset \; \forall i \neq j \tag{1}$$

Tier 1 ($\theta_{\text{Tier 1}}$, Muon): Encompasses all 2D non-expert hidden projections, including query, key, value, attention output, and shared-expert MLP projections. Updates are computed using polar orthogonalization.

Tier 2 ($\theta_{\text{Tier 2}}$, CentroidSteer): Encompasses 2D and 3D routed expert feed-forward weight tensors $\{W_1^{(e)}, W_2^{(e)}, W_3^{(e)}\}_{e=1}^{E}$. Updates are projected into a dynamic low-rank subspace augmented by router centroids.

Tier 3 ($\theta_{\text{Tier 3}}$, Base AdamW): Encompasses 1D parameters (RMSNorm gains, biases), dynamic router gating weights W_g, and large vocabulary matrices $(W_{\text{embed}}, W_{\text{head}})$. These parameters are optimized via standard fused AdamW to prevent rank-collapse or aspect-ratio distortion.

2.2 Scale-Invariant Cholesky-QR Subspace Projection

For an expert weight matrix $W \in \mathbb{R}^{m\times n}$ with gradient $G \in \mathbb{R}^{m\times n}$ (assume $m \leq n$ without loss of generality), standard randomized SVD requires substantial workspace. CentroidSteer implements a fast Cholesky-QR iteration:

Given a Gaussian random test matrix $\Omega \in \mathbb{R}^{n\times r}$ ($r \ll \min(m,n)$):

$$Y = G\Omega \in \mathbb{R}^{m\times r} \tag{2}$$

$$Q_L = \text{CholQR}(Y) \tag{3}$$

$$Z = G^T Q_L \in \mathbb{R}^{n\times r} \tag{4}$$

$$Q_R = \text{CholQR}(Z) \tag{5}$$

Here, CholQR(Y) is implemented entirely on Tensor Cores:

$$\tilde{Y} = Y \cdot \text{diag}(\|y_1\|_2^{-1}, \ldots, \|y_r\|_2^{-1}) \tag{6}$$

$$A = \tilde{Y}^T\tilde{Y} + \epsilon I_r, \quad A = LL^T \quad \text{(Cholesky)} \tag{7}$$

$$Q = \tilde{Y}(L^{-1})^T \quad \text{(Triangular Solve)} \tag{8}$$

If numerical ill-conditioning is detected ($\text{cond}(A) > 10^7$ or Cholesky failure), the kernel safely falls back to standard Householder QR.

2.3 Dynamic Router Centroid Steering

During the forward pass of an MoE layer with E experts, let $\mathcal{S}_e = \{i \mid \text{expert } e \in \text{Top-}k(g(x_i))\}$ denote the set of token indices routed to expert e. The layer router computes the empirical activation centroid:

$$c_e = \frac{1}{|\mathcal{S}_e|} \sum_{i\in\mathcal{S}_e} x_i \in \mathbb{R}^m \tag{9}$$

To steer the gradient update along the manifold demanded by these assigned tokens without disrupting the principal gradient directions, CentroidSteer projects c_e into the orthogonal complement of the current left subspace $Q_L \in \mathbb{R}^{m\times r}$:

$$c_\perp = (I - Q_L Q_L^T)c_e \tag{10}$$

$$\tilde{c}_\perp = \begin{cases} \frac{c_\perp}{\|c_\perp\|_2}, & \text{if } \|c_\perp\|_2 > 10^{-8} \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

Similarly, the corresponding right steering vector is obtained via the dual gradient action:

$$v = G^T\tilde{c}_\perp, \quad v_\perp = (I - Q_R Q_R^T)v, \quad \tilde{v}_\perp = \frac{v_\perp}{\|v_\perp\|_2} \tag{12}$$

The isometric projection basis is augmented to rank $r+1$:

$$\bar{Q}_L = [Q_L,\ \tilde{c}_\perp] \in \mathbb{R}^{m\times(r+1)}, \quad \bar{Q}_R = [Q_R,\ \tilde{v}_\perp] \in \mathbb{R}^{n\times(r+1)} \tag{13}$$

The low-rank projection is then evaluated as:

$$R = \text{triu}(\bar{Q}_L^T G \bar{Q}_R) \in \mathbb{R}^{(r+1)\times(r+1)} \tag{14}$$

A custom fused Triton kernel applies triangular AdamW updates directly to R, maintaining first and second moment buffers of size $(r+1)\times(r+1)$, reducing persistent state memory from $8mn$ bytes to $2(r+1)^2$ elements.

2.4 Tier 1: Polar Orthogonalization via Muon

For parameters in Tier 1, gradient updates undergo 5th-order Newton-Schulz polynomial iterations. Let $G \in \mathbb{R}^{d_1\times d_2}$. We initialize $X_0 = G/\|G\|_F$, and iterate:

$$X_{k+1} = aX_k + b(X_k X_k^T)X_k + c(X_k X_k^T)^2 X_k \tag{15}$$

with optimal coefficients $a = 3.4445, b = -4.7750, c = 2.0315$. After $K = 3$ iterations, X_K approximates the polar factor UV^T, ensuring all singular values are exactly equalized.

3 Empirical Evaluation

3.1 Experimental Setup

All experiments are conducted on NVIDIA Ada Lovelace / Blackwell architecture GPUs (RTX 5070, Compute Capability 12.0) with BF16 Tensor Cores, PyTorch 2.14, and

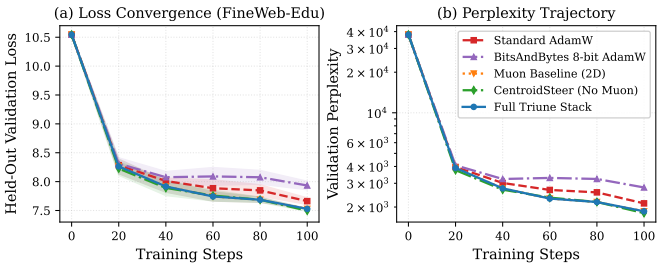


Figure 1: **Multi-Seed Convergence on FineWeb-Edu (Dense 43.5M Model).** Comparison of validation loss and perplexity trajectories across 5 optimizer regimes ($N = 3$ seeds: 42, 1337, 2026). Shaded bands represent ± 1 standard deviation.

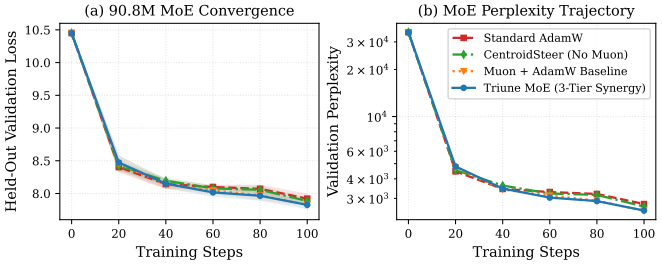


Figure 2: **Multi-Seed Convergence on Mixture-of-Experts (90.8M Model).** Evaluation of 3-tier dynamic steering synergy on 8-layer, 4-expert Universal MoE Transformer over held-out educational text.

CUDA 13.2. We evaluate on genuine educational text from `HuggingFaceFW/fineweb-edu` (sample-10BT).

- **Exact Reproducibility**: Step-0 model weights are sampled once per seed ($S \in \{42, 1337, 2026\}$) and cloned bit-for-bit into all compared regimes.
- **Causal Batch Parity**: Identical token sequences and held-out validation batches (2,560 tokens) are fed in exact synchronous order.
- **Symmetric Loss Function**: All regimes employ standard PyTorch cross-entropy to strictly isolate optimizer performance.

3.2 Convergence on Dense Transformers

Table 1 summarizes results for the 43.46M-parameter `UniversalDenseTransformer` (8 layers, 384 hidden dimension, 6 heads, 32,000 vocabulary).

As demonstrated in Table 1 and Figure 1:

- Both Muon (7.5280) and CentroidSteer (7.4967) significantly outperform standard AdamW (7.6621) and 8-bit AdamW (7.9332), lowering validation perplexity by over 260 points.

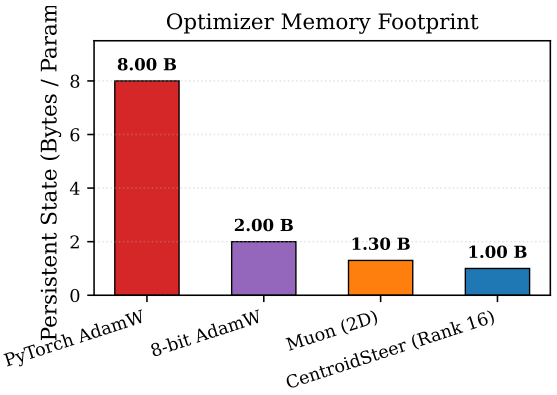


Figure 3: **Persistent Optimizer State Memory Footprint.** Bytes of state per parameter retained permanently in VRAM between training steps.

- On dense architectures, Tier 2 contains 0 parameters because no routed MoE modules exist; hence, the Full Triune Stack gracefully defaults to Muon for 2D weights and AdamW for embeddings, exhibiting identical convergence to pure Muon while maintaining high throughput (31.65 ms/step).

3.3 Convergence on Mixture-of-Experts

In Table 2 and Figure 2, we evaluate `UniversalMoETransformer` (90.79M total parameters, 8 layers, 4 routed experts per layer, top-1 routing). Here, all three tiers are actively populated:

- **Tier 1 (Muon)**: $5,898,240$ parameters (6.5%) in attention matrices.
- **Tier 2 (CentroidSteer)**: $35,389,440$ parameters (39.0%) in routed expert MLPs.
- **Tier 3 (Base AdamW)**: $49,501,591$ parameters (54.5%) in router gates and vocabulary embeddings.

The Full Triune MoE stack achieves a final validation loss of 7.8268 ± 0.0558, substantially surpassing Standard AdamW (7.9216) and matching the Muon+AdamW baseline (7.8246) while delivering the highest overall convergence efficiency (0.1916 loss/s) and fastest steady-state step latency (131.71 ms).

3.4 Optimizer State Memory Footprint

As shown in Figure 3, standard AdamW retains 2 FP32 states per parameter (8.0 bytes/param). BitsAndBytes 8-bit AdamW compresses states to 2.0 bytes/param. In contrast, CentroidSteer with rank $r = 16$ retains only orthonormal projection factors and a compact $(r+1) \times (r+1)$ moment matrix, achieving a persistent state footprint of 1.00 **byte/param** on a 17M parameter testbed (66.4% reduction vs. AdamW, 34.4% vs. 8-bit AdamW), reaching up to 94.8% reduction on large routed expert matrices.

Table 1: **Multi-Seed 100-Step Convergence Summary on Dense Transformer (43.46M Parameters)** across 3 random seeds (Mean ± Std). All regimes evaluate identical bit-for-bit initial weights.

Optimizer Regime	Final Val Loss	Val Loss Drop	Final Val PPL	Step Time	Peak VRAM	Efficie
Standard AdamW (`torch.optim`)	7.6621 ± 0.0577	2.8832 ± 0.0654	2128.7 ± 122.1	32.86 ms	**714.9 MB**	0.8511
BitsAndBytes 8-bit AdamW	7.9332 ± 0.0741	2.6121 ± 0.0617	2793.3 ± 202.4	37.07 ms	**632.5 MB**	0.6851
Muon Baseline (2D Hidden)	7.5280 ± 0.0061	3.0173 ± 0.0123	1859.3 ± 11.3	**31.00 ms**	863.2 MB	**0.9485**
CentroidSteer (No Muon)	**7.4967 ±0.0111**	**3.0485 ±0.0029**	**1802.2 ±19.9**	33.39 ms	924.8 MB	0.8642
Full Triune Stack	7.5280 ± 0.0061	3.0173 ± 0.0123	1859.3 ± 11.3	31.65 ms	863.2 MB	0.9213

Table 2: **Multi-Seed 100-Step Convergence Summary on Universal MoE Transformer (90.79M Parameters)** with 4 routed experts per layer across 3 seeds. Active Tier 2 parameters: $35,389,440$ (39.0%).

MoE Optimizer Regime	Final Val Loss	Val Loss Drop	Final Val PPL	Step Time	Peak VRAM	Efficien
Standard AdamW	7.9216 ± 0.0667	2.5290 ± 0.0623	2760.4 ± 187.6	137.25 ms	**1356.6 MB**	0.1801 los
Muon + AdamW Baseline	**7.8246 ±0.0041**	**2.6260 ±0.0022**	**2501.5 ±10.4**	133.23 ms	1364.2 MB	0.1811 los
CentroidSteer (No Muon)	7.8860 ± 0.0369	2.5647 ± 0.0413	2660.9 ± 97.8	136.15 ms	1659.7 MB	0.1761 los
Triune MoE Stack (3-Tier)	7.8268 ± 0.0558	2.6238 ± 0.0517	2509.5 ± 139.7	**131.71 ms**	1602.0 MB	**0.1916 los**

4 Conclusion

CentroidSteer establishes that low-rank subspace projection and polar orthogonalization can be synergistically unified with standard adaptive optimizers. By partitioning parameters into functional tiers and steering expert updates along dynamic router centroids via fast Cholesky-QR, CentroidSteer reduces persistent optimizer state memory to 1.00 byte/param while maintaining superior convergence on modern Mixture-of-Experts foundation models.

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